

FIBRE FORWARD: UNLOCKING THE POWER OF FIBRE IN MANAGING GI HEALTH

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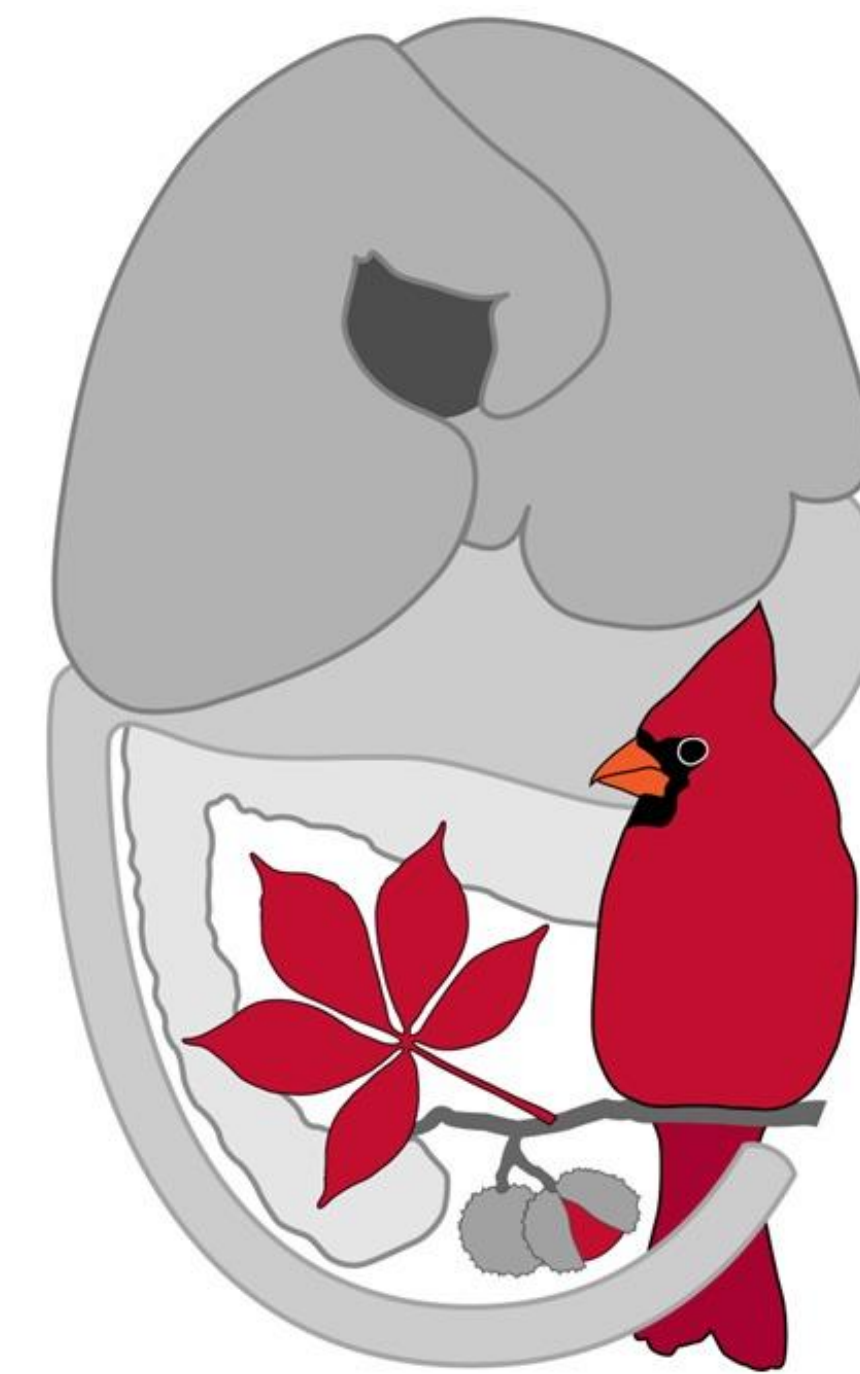
Education



- BS, The Ohio State University (Biology and Ecology)
- DVM, The Ohio State University (Small Animal Focus)
- Internship, Purdue University (Small Animal Medicine and Surgery)
- Residency, The Ohio State University (Diplomate American College of Veterinary Internal Medicine)
- MS, The Ohio State University (Comparative Veterinary Medicine – Enteroendocrine Physiology)
- PostDoc, The Ohio State University (Veterinary Biosciences – Mucosal Immunology)
- Post Doc, Nationwide Childrens Hospital (Comparative Medicine – Microbial Pathogenesis)



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CHIRP

Comparative Hepatobiliary and Intestinal research Program

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Friends and Family





So, What is Dietary Fiber?



Simplest Definition:

Dietary fiber includes a diverse assortment of non-digestible carbohydrates.



Functional Properties



Fermentability is defined by the speed and completeness for which a given fiber type can be fermented by microbes in the GIT.

Solubility is a measure of their ability to be suspended in water.

Viscosity is a measure of the fiber types ability to thicken and form gels in solution.



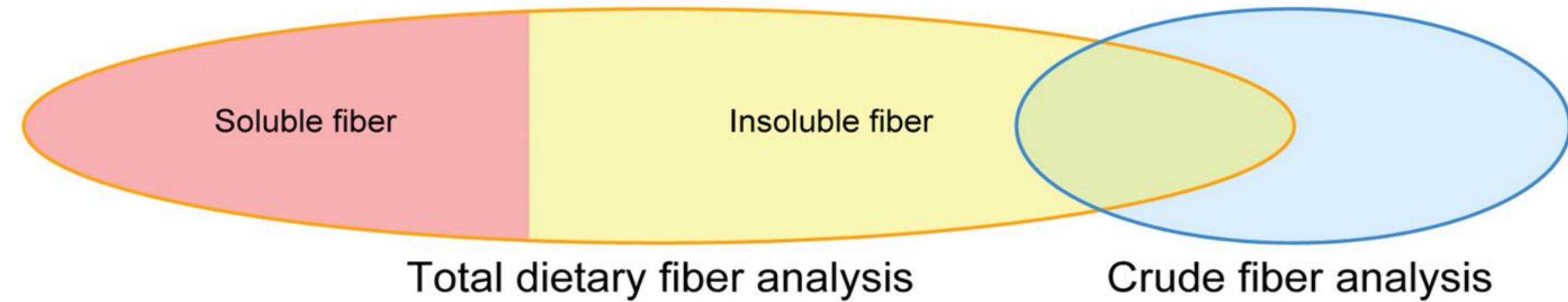


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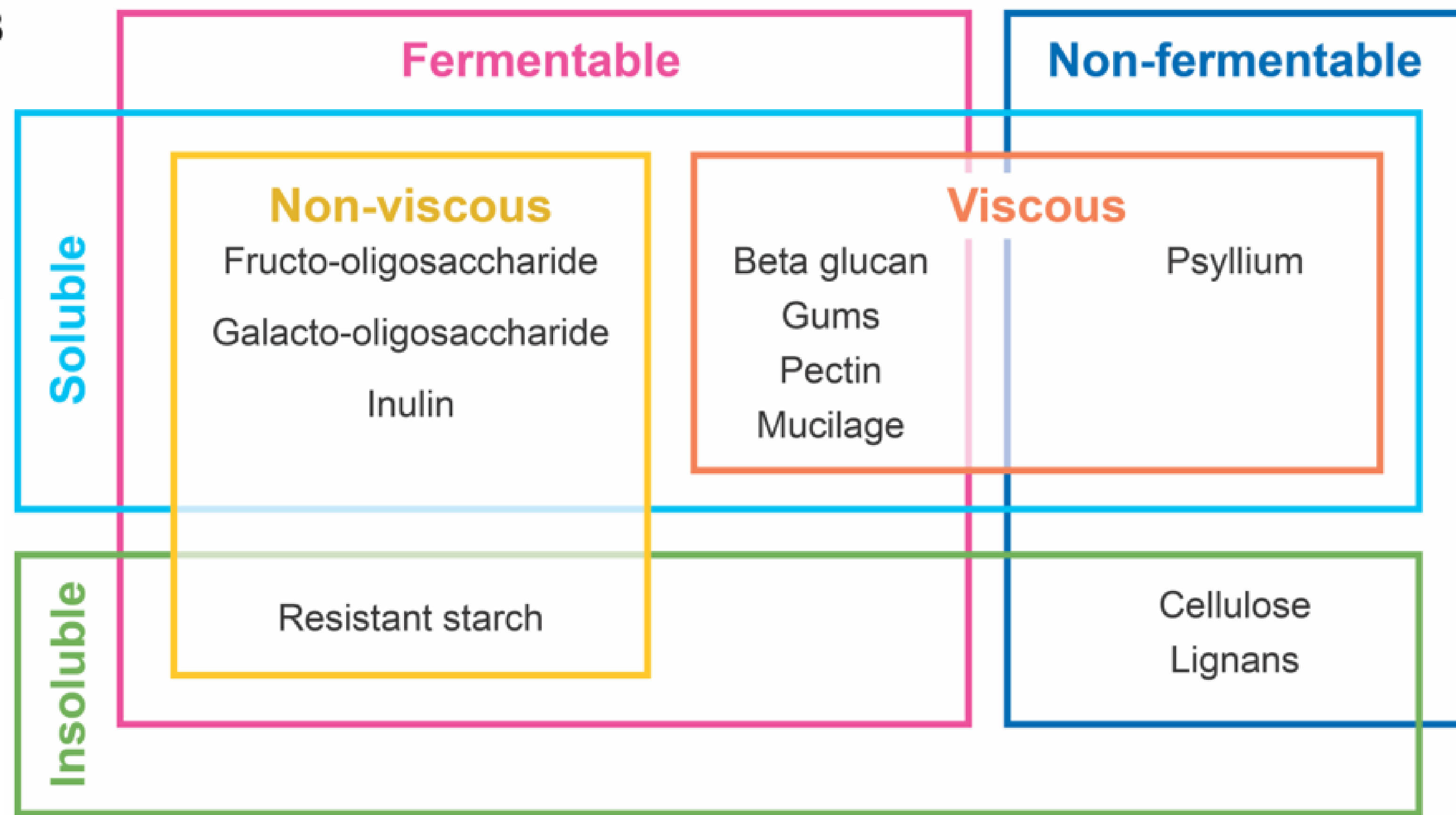
Dietary Fiber Sources		Solubility		
		High	Moderate	Low
Fermentability	High	Fructose, Galactans, Mannans, Mucilages, Apple pectin, Citrus pulp, Guar gum, Gum arabic, Soy fiber, Apple pomace, Carrot pomace, Citrus pectin	Psyllium	
	Moderate	Pectin, Flaxseed	Grape pomace, Tomato pomace, Pea hulls	Hemicellulose, Beet pulp, Corn bran, Pea fiber, Rice bran, Soy hulls, Wheat brans, Wheat middlings
	Low		Pistachio	Hemicellulose, Cellulose, Soy hulls, Peanut hulls, Sunflower hulls

Rudinsky AJ and Parker VJ. VCNA. 2026.

B



B



Rudinsky AJ and Parker VJ. VCNA. 2026.

General Rules of Fermentation

as the Fermentation Rate of Fiber INCREASES

Gastrointestinal transit time decreases
Fecal bulk decreases
Fecal metabolite production increases

as the Fermentation Rate of Fiber DECREASES

Gastrointestinal transit time increases
Fecal bulk increases
Fecal metabolite production decreases.

Lewis LD, et al. J Nutr. 1994.
Bueno L, et al. Gastroenterology. 1981.
Roberfroid M. Crit Rev Food Sci Nutr. 1993.





Why Should We All Care about Dietary Fiber?



As a Gastroenterologist ?

Dietary fiber plays a vital role in the health of animals and maintenance of gastrointestinal tract (GIT) homeostasis.

- Physically altering the digesta
- Modulating appetite and satiety
- Regulating digestion
- Acting as a microbial energy source through fermentation

Moreno A, et al. 2022.

Proposed Benefits

- Cardiovascular Health
- Reproductive Health
- Decreased Neoplasia Risk

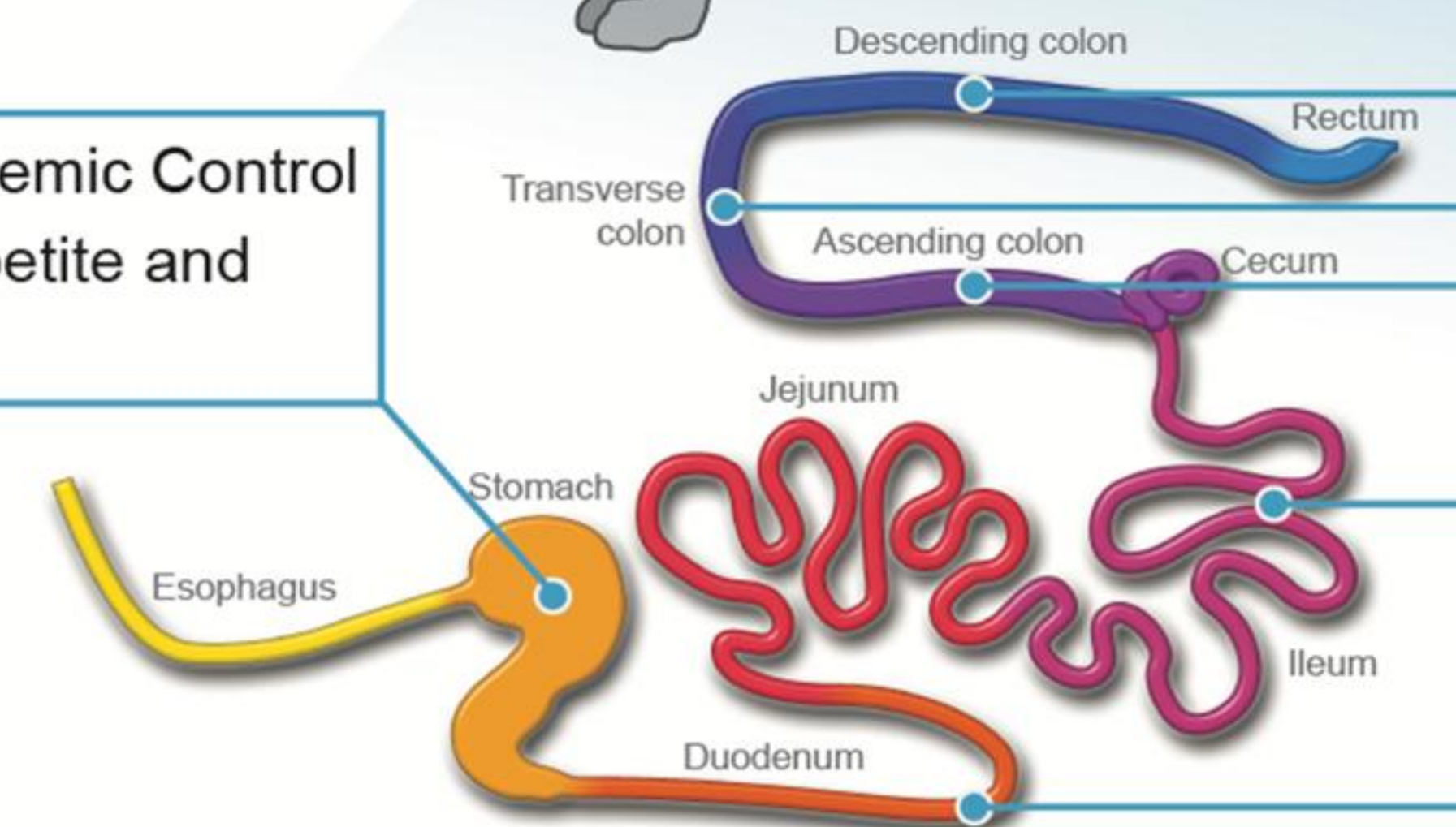
Improved Immune System Health
(e.g. IgA)

- Prebiotic
- Short Chain Fatty Acid Production
- Alter Microbiome

- Increase Fecal Volume
- Increased Bowel Movement Frequency
- Alter Fecal Consistency and Moisture Content

- Improve Glycemic Control
- Regulate Appetite and Body Weight

- Regulate GIT Transit Time
- Affect Nutrient Digestibility
- Alter Nutrient Availability

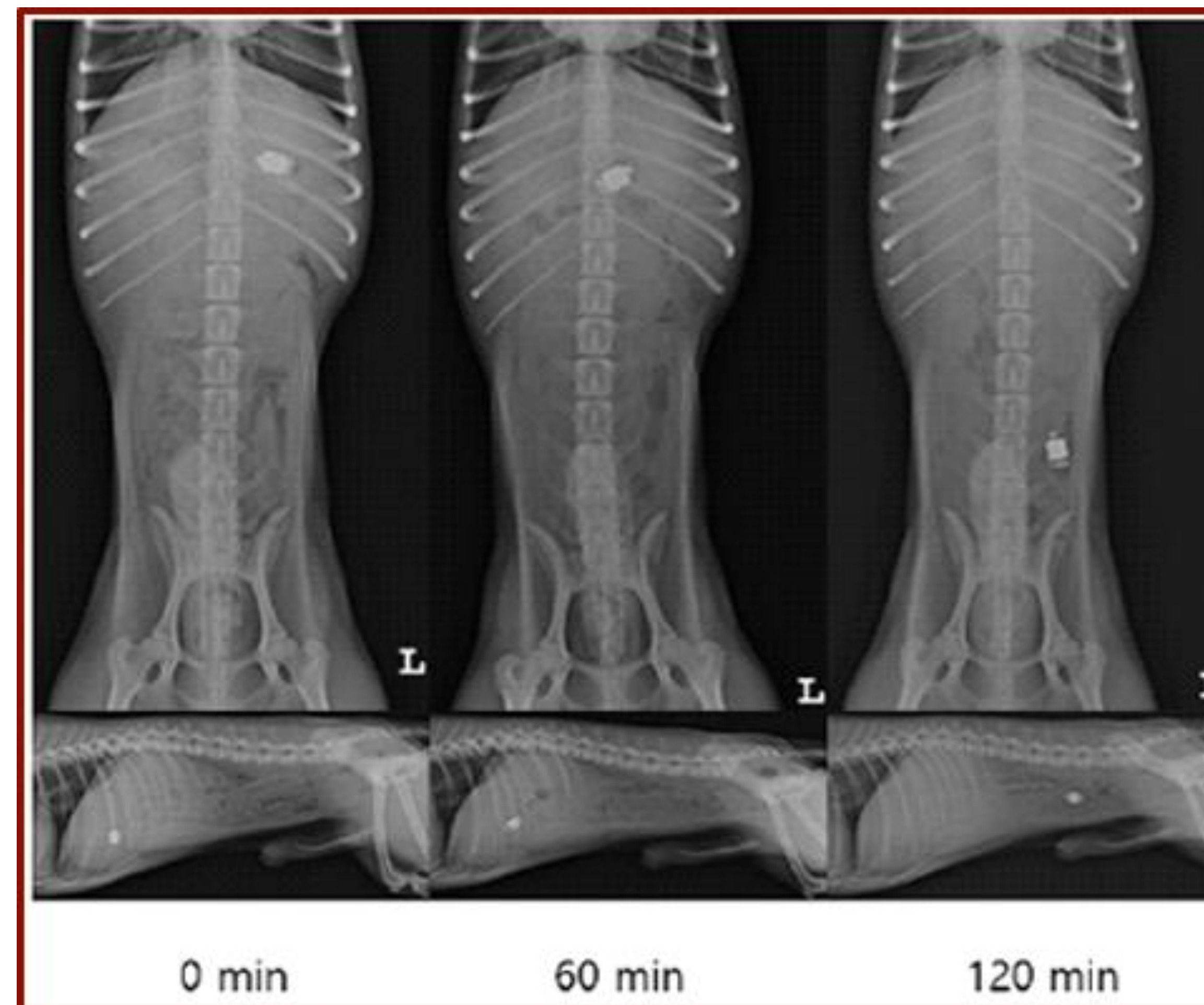


Rudinsky AJ, et al. 2023.

Gastrointestinal Transit Time

Fiber helps to regulate stool transit time through:

- Bulking effects
- Regulating GIT water concentration
- Stimulating mucus secretion
- Bacterial metabolism



Lewis LD, et al. J Nutr. 1994.
Bueno L, et al. Gastroenterology. 1981.
Roberfroid M. Crit Rev Food Sci Nutr. 1993.



Gastrointestinal Transit Time



Fiber solubility and viscosity provide the most consistent predictive effect of the impact on GIT transit times

Soluble and Viscous Fiber

- Increase gastric emptying time
- Increase small intestinal transit time

Insoluble and Nonviscous Fibers

- Increase gastric emptying time
- Decrease transit time in both the small intestine and colon

Lewis LD, et al. J Nutr. 1994.
Bueno L, et al. Gastroenterology. 1981.
Roberfroid M. Crit Rev Food Sci Nutr. 1993.





KEY POINTS

The way to alter and prescribe diet is very different depending on whether you are dealing with vomiting, small bowel diarrhea or large bowel diarrhea.



Laxation, Water Content, and Volume



Soluble, Fermentable Fibers

- Increase stool weight, moisture concentration, and softness
- This occurs through increasing water retention, fecal dry matter percentage, and bacterial concentration

Insoluble, Minimally Fermentable Fibers

- Increase fecal mass
- This occurs through water retention and its dry matter bulking capacity

Lewis LD, et al. J Nutr. 1994.
Bueno L, et al. Gastroenterology. 1981.
Roberfroid M. Crit Rev Food Sci Nutr. 1993.



KEY POINTS

There is evidence in dogs that defecation frequency, fecal score, and moisture content may be altered by modifying dietary fiber, with minimal supporting evidence in cats.

The effects on fecal score and fecal appearance can be particularly useful as abnormalities in fecal appearance, whether clinical or not, can be distressing for owners.

KEY POINTS



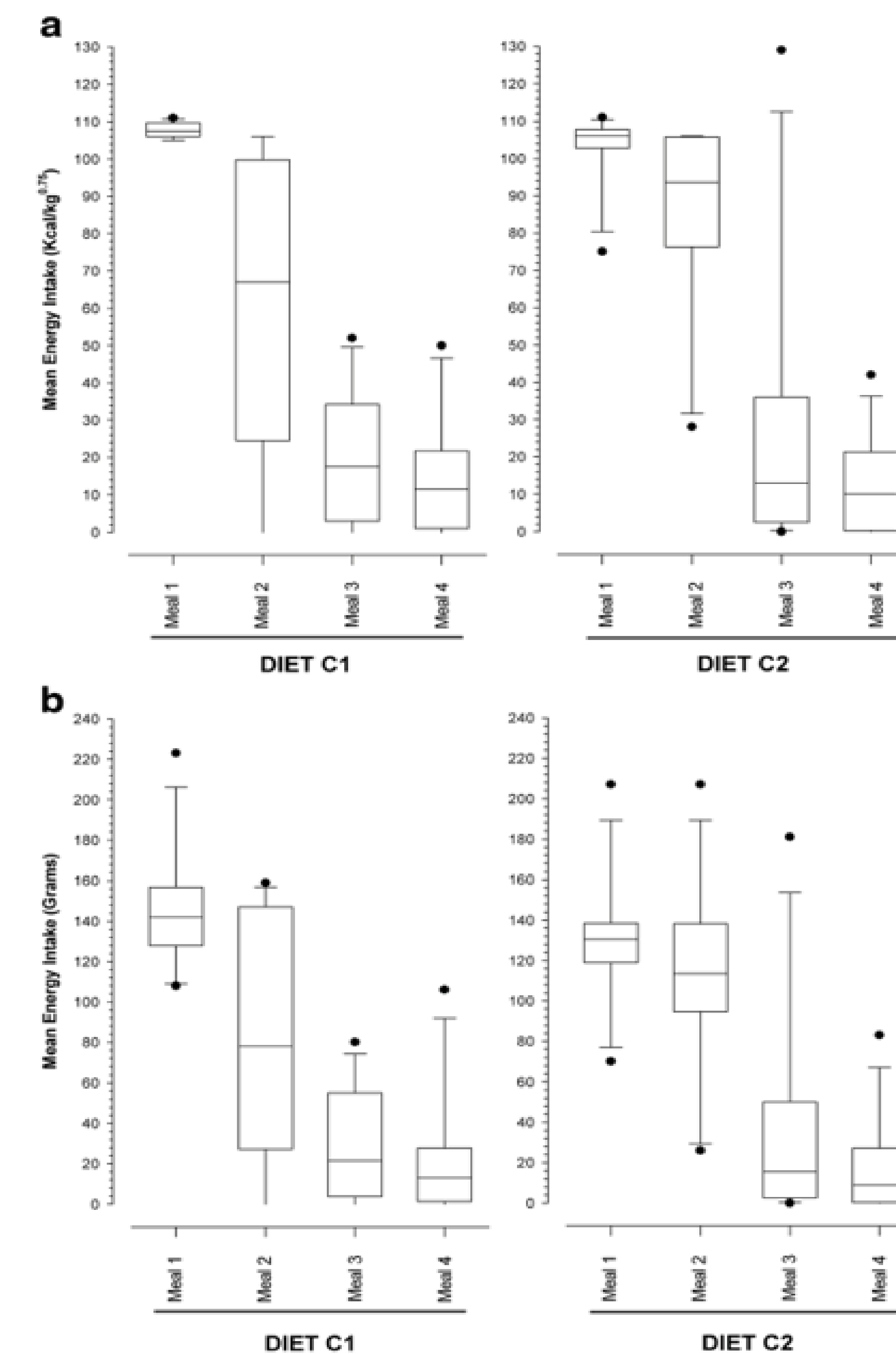
Appetite and Body Weight

Increasing dietary fiber has long been a staple in weight management therapeutic strategies

- Decrease caloric density of foods
- Alter dry and organic matter digestibility
- Increase satiety



Rudinsky and Parker. VCNA. 2026.



Hours et al. BMC Veterinary Research. 2016.



KEY POINTS

- **There is evidence in both dogs and cats that reducing the caloric content of food through the addition of dietary fiber is beneficial in weight loss programs.**
- **Despite palatability concerns with higher fiber foods in human nutrition, this dose not seem to be an issue in dogs.**



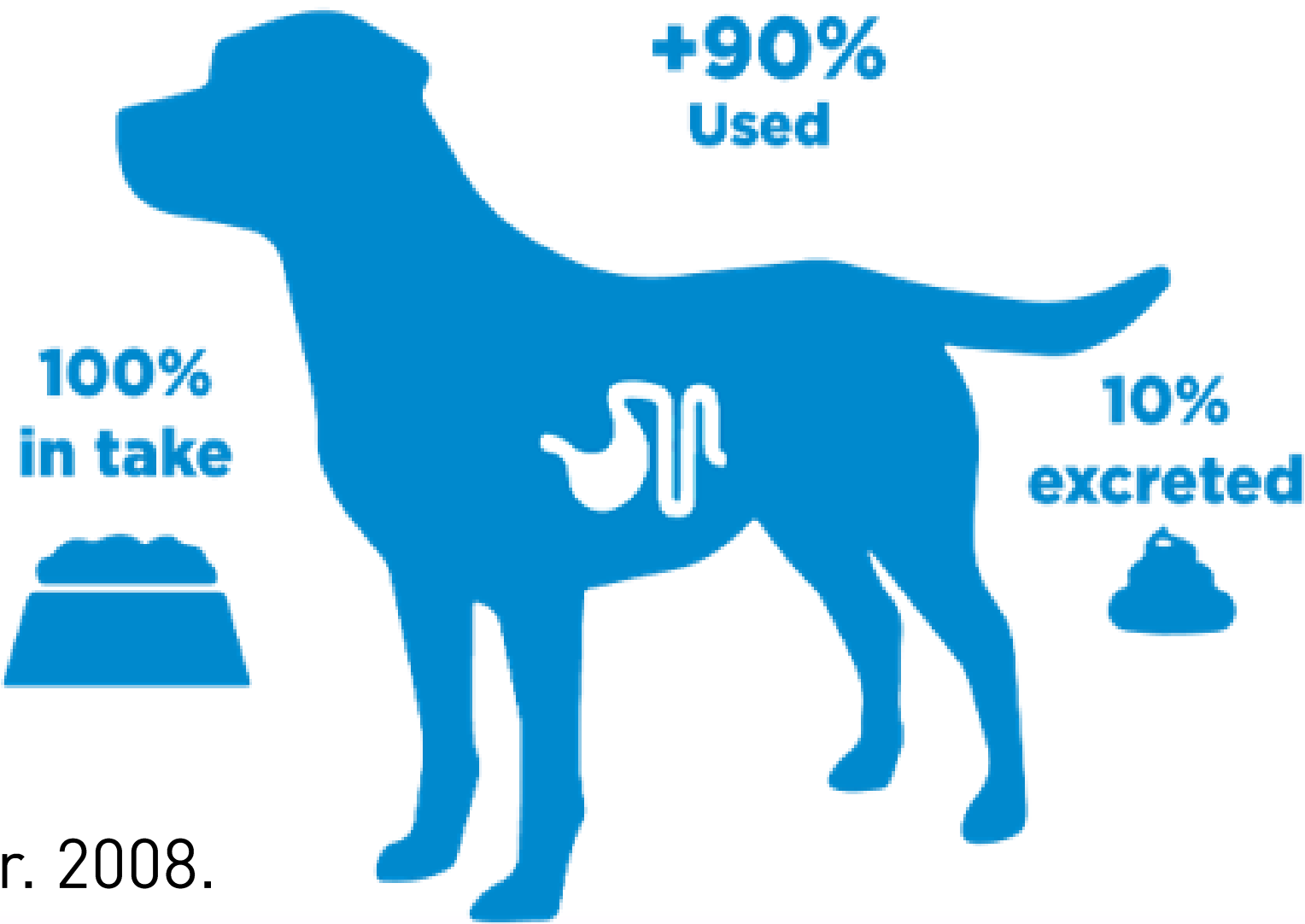
Digestibility

Fiber decreases the overall digestibility of a diet

- Slowly fermentable fibers cause a greater reduction
- Soluble fibers alter nutrient availability by forming gels and acting as a physical barrier to enzymatic digestion
- Soluble fibers modulate GIT transit time affecting enzyme-substrate interactions

Item	Cassava flour	Corn	Sorghum	Brewer's rice	Lentil	Pea
<i>Daily nutrient intake (g/kg body weight /day)</i>						
Dry matter	16.3 ± 0.5	15.2 ± 1.5	14.3 ± 0.5	15.2 ± 1.3	18.9 ± 0.5	16.5 ± 0.2
Organic matter	15.0 ± 0.5	13.9 ± 1.4	13.2 ± 0.4	13.9 ± 1.2	17.4 ± 0.5	15.1 ± 0.2
Crude protein	5.0 ^{bc} ± 0.2	4.8 ^{bc} ± 0.5	4.1 ^c ± 0.1	5.6 ^{bc} ± 0.5	5.9 ^b ± 0.2	4.9 ^{bc} ± 0.1
Starch	6.8 ± 0.2	5.8 ± 0.6	5.5 ± 0.2	6.1 ± 0.5	7.3 ± 0.2	6.0 ± 0.1
Fat	2.1 ^b ± 0.1	1.8 ^{bc} ± 0.2	1.4 ^c ± 0.1	1.7 ^{bc} ± 0.1	1.7 ^{bc} ± 0.1	1.8 ^{bc} ± 0.1
Total dietary fibre	1.0 ^{de} ± 0.1	1.4 ^{cd} ± 0.1	1.7 ^c ± 0.1	0.9 ^e ± 0.1	3.0 ^b ± 0.1	2.6 ^b ± 0.1
Crude fibre	1.9 ± 0.2	2.5 ± 0.2	2.0 ± 0.1	2.3 ± 0.2	2.5 ± 0.1	2.4 ± 0.1
<i>Apparent digestibility coefficients (%)</i>						
Dry matter	83.1 ^b ± 0.1	78.6 ^c ± 0.5	79.0 ^c ± 0.3	82.4 ^b ± 0.	74.5 ^d ± 0.6	76.1 ^d ± 0.7
Organic matter	87.7 ^b ± 0.1	83.9 ^c ± 0.4	83.8 ^c ± 0.2	88.4 ^b ± 0.2	79.3 ^d ± 0.4	80.1 ^d ± 0.7
Crude protein	86.8 ^{bc} ± 0.5	86.1 ^c ± 0.3	85.0 ^{cd} ± 0.5	89.0 ^b ± 0.5	79.9 ^e ± 0.7	83.4 ^d ± 0.6
Starch	99.4 ^b ± 0.1	99.1 ^{bc} ± 0.1	99.1 ^{bc} ± 0.1	99.3 ^b ± 0.1	98.8 ^c ± 0.2	98.7 ^c ± 0.1
Fat	92.8 ^b ± 0.2	89.1 ^{cd} ± 0.3	88.3 ^d ± 0.1	89.0 ^{cd} ± 0.4	89.4 ^{cd} ± 0.5	90.0 ^c ± 0.2
Gross energy	87.8 ^b ± 0.1	84.9 ^c ± 0.6	84.2 ^c ± 0.4	87.7 ^b ± 0.4	78.4 ^e ± 0.5	80.9 ^d ± 0.8
Total dietary fibre	8.2 ^d ± 2.2	11.4 ^d ± 3.3	27.0 ^{bc} ± 1.4	17.7 ^{cd} ± 0.7	33.4 ^b ± 1.7	30.3 ^b ± 3.0
Crude fibre	2.6 ± 0.7	3.1 ± 0.3	3.7 ± 0.3	2.6 ± 0.3	5.0 ± 1.2	4.0 ± 0.8
<i>Faecal characteristics</i>						
Faecal score†	4.2 ± 0.1	3.7 ± 0.1	3.8 ± 0.1	4.2 ± 0.1	3.7 ± 0.1	3.7 ± 0.1
Faecal DM (%)	38.6 ^{cd} ± 1.9	40.2 ^{bc} ± 1.2	40.3 ^{bc} ± 1.6	48.5 ^b ± 2.6	33.9 ^{cd} ± 2.7	30.9 ^d ± 1.4
pH faecal	6.7 ^{bc} ± 0.1	6.5 ^{bcd} ± 0.1	6.7 ^{bc} ± 0.1	7.1 ^b ± 0.1	6.0 ^d ± 0.1	6.2 ^{cd} ± 0.2

*Values are means ± SE of the six dogs per diet.
^{bcd} Within a row, means without a common superscript differ (p < 0.05).



De-Oliveira LD, et al. J Anim Physiol Anim Nutr. 2008.



Pertinent Feline Studies



Comparison study of GI transit
time in healthy young versus
geriatric

Conclusion:
Slower transit in geriatric cats

Comparison study of macronutrient
digestibility of healthy young
versus geriatric

Conclusion:
reduced digestibility of protein, fat,
and carbohydrates in geriatric cats



KEY POINTS

**Consider the fiber effects in scenarios
of proximal gastrointestinal disease and decreased
functional capacity of the gastrointestinal tract.**



Prebiotics



“The definition of a prebiotic fiber set forth by the International Scientific Association of Probiotics and Prebiotics is a **substrate** selectively **used by host microorganisms** which confers a **health benefit.**”

“Fibers are classified as prebiotics if they are resistant to digestion by host enzymes, promote the growth of symbiotic gut microbe, and fail to promote the growth of pathogenic bacteria.”

Hill, C, et al. Nature Reviews Gastroenterology and Hepatology. 2014.



What is happening?

Fibers are primarily broken down by *Bifidobacterium spp.* and *Lactobacillus spp.* And remain inaccessible to other gut microbes in dogs and cats as well

Increased consumption of these carbohydrates gives a competitive advantage to and promotes the growth of these bacteria

Confer host benefits locally within the GIT and systemically



Prebiotic factor	Origin	Microbes stimulated	Effects
Oligosaccharides	Onion,garlic, chicory root, burdock, asparagus, Jerusalem artichoke, soybean, wheat bran.	<i>Bifidobacterium</i> species	Increase in bifidobacterium, suppression of putrefactive bacteria, prevention of constipation and diarrhea.
Fructooligosaccharides (inulin, oligofructo)	Same as for oligosaccharides	<i>Bifidobacterium</i> species <i>Lactobacillus acidophilus</i> , <i>Lactobacillus casei</i> , <i>Lactobacillus plantarum</i>	Growth of bifidobacteria and acid promotion.
Fructan	Ash-free white powder from tubers of Jerusalem artichoke.	<i>Bifidobacterium</i> species	Growth of bifidobacteria
Human kappa casein and derived glycolmacropeptide	Human milk: chymotrypsin and pepsin hydrolysate.	<i>Bifidobacterium bifidum</i>	Growth promotion.
Stachyose and raffinose	Soybean extract	<i>Bifidobacterium</i> species	Growth factor.
Casein macropeptide	Bovine milk	<i>Bifidobacterium</i> species	Growth promotion.
Lactitol(4-O-β-D-galactopyranosyl)D-glucitol	Synthetic sugar alcohol of lactose	<i>Bifidobacterium</i> species	Growth promotion.
Lactutose(4-O-β-D-galactopyranosyl)D-fructose	Synthetic derivative of lactose	<i>Bifidobacterium</i> species	Growth promotion.





TABLE IV. Faecal microbial concentrations [$\log_{10}$ cfu/g dry faeces] for senior dogs fed a control diet (CON) or a diet supplemented with chicory (CH), mannan-oligosaccharides (MOS), or a combination (CM)¹

Item	Treatment				SEM	Control vs. oligosaccharide Contrast P value		
	CON	CH	MOS	CM		CH	MOS	CM
Bifidobacteria	10.1	10.5	10.6	10.3	0.14	0.05	0.03	NS ²
Lactobacilli ³	9.1	10.1	9.6	9.6	0.19	NS	NS	NS
<i>E. coli</i>	7.6	7.5	6.2	7.7	0.38	NS	0.04	NS
<i>Clostridium perfringens</i>	9.1	9.3	9.0	8.9	0.15	NS	NS	NS

¹Values are least squares means.

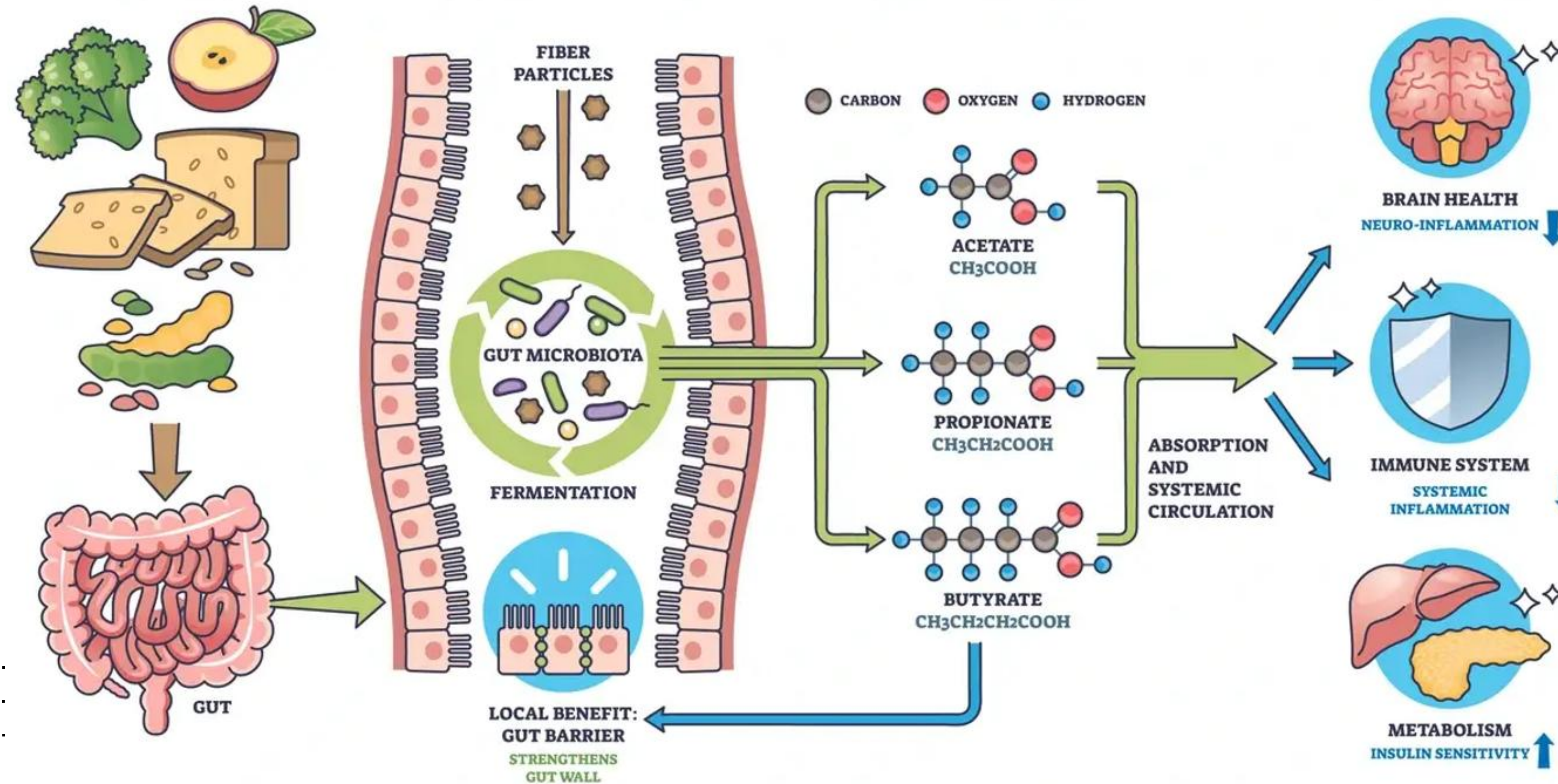
²NS = not significantly different ($P \geq 0.15$).

³The treatment by block interaction was significant in analysis of faecal lactobacilli concentrations. Therefore, this term was left in the model for analysis of this variable only.

“Supplementation with either MOS or chicory (inulin) in dogs increased fecal concentration of *Bifidobacterium spp.*”



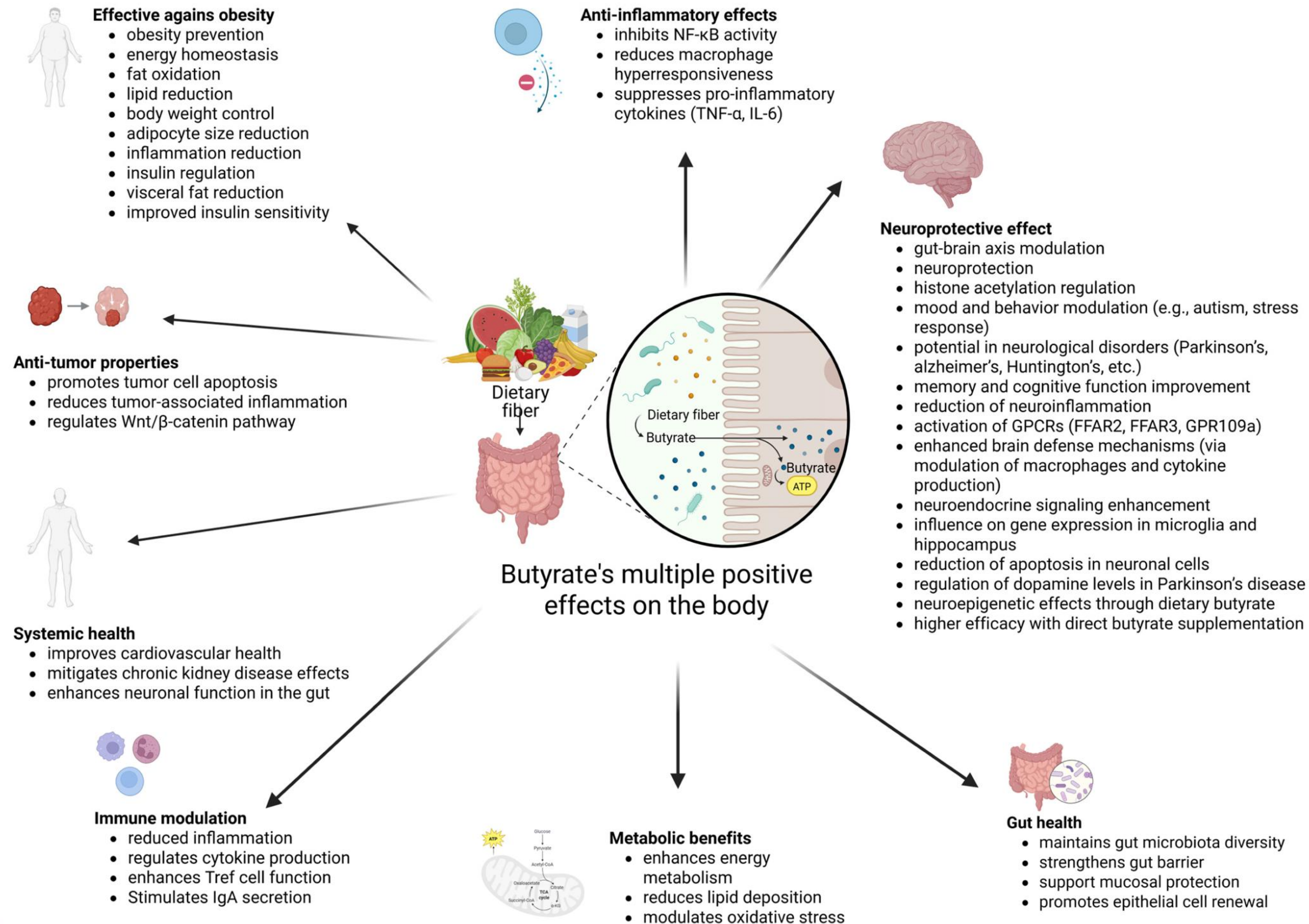
Fermentation results in the production of metabolites [i.e. short chain fatty acids (SCFA)]



Tan J, et al. Adv Immunol. 2014.
Kiely EM, et al. J pediatric Surg. 2001.
Scheppach W, et al. Gastroenterology. 1992.



Butyrate



Kalka AE, et al. Nutrients. 2025.





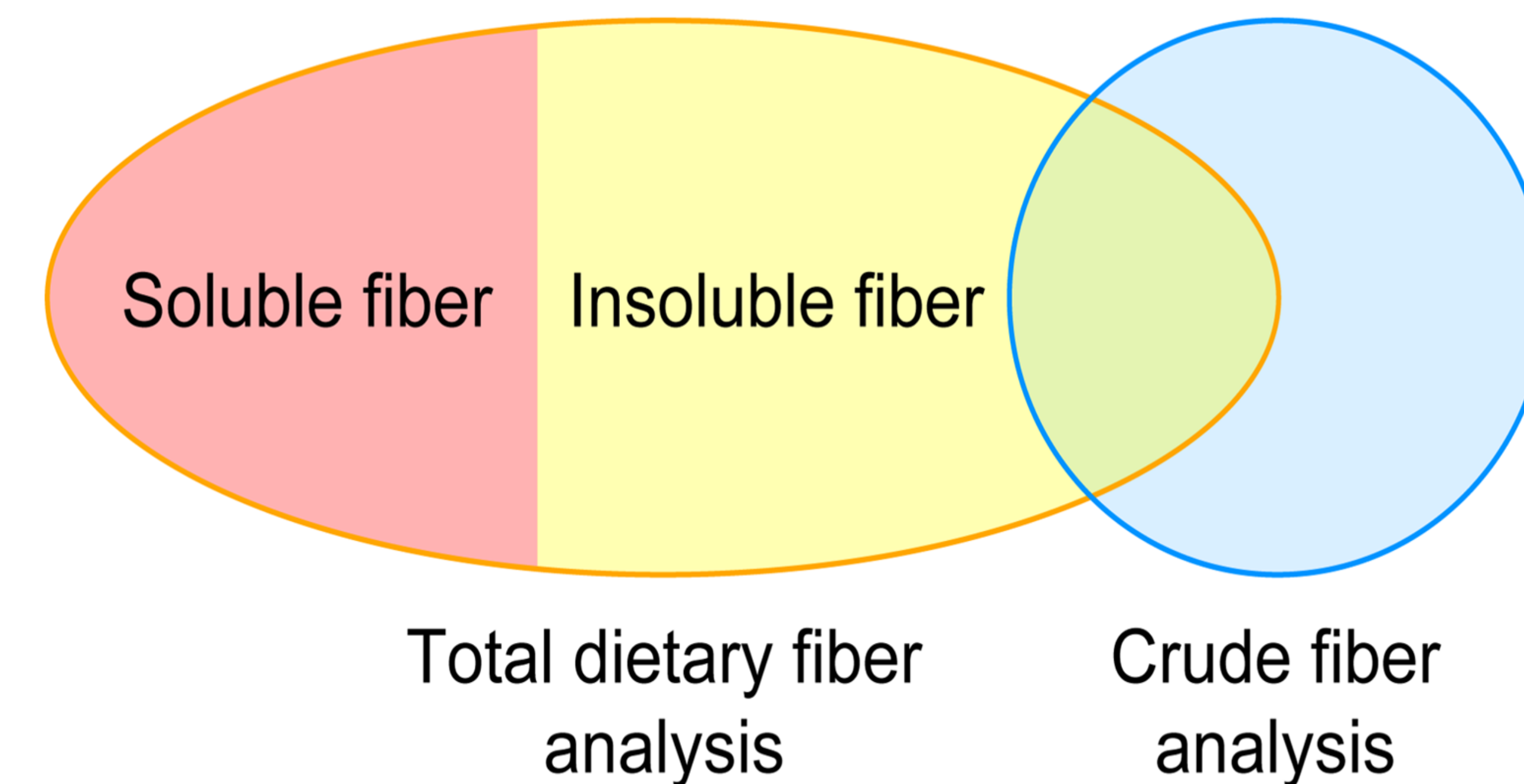
Dietary Fibre in Pet Foods



Fiber Reporting



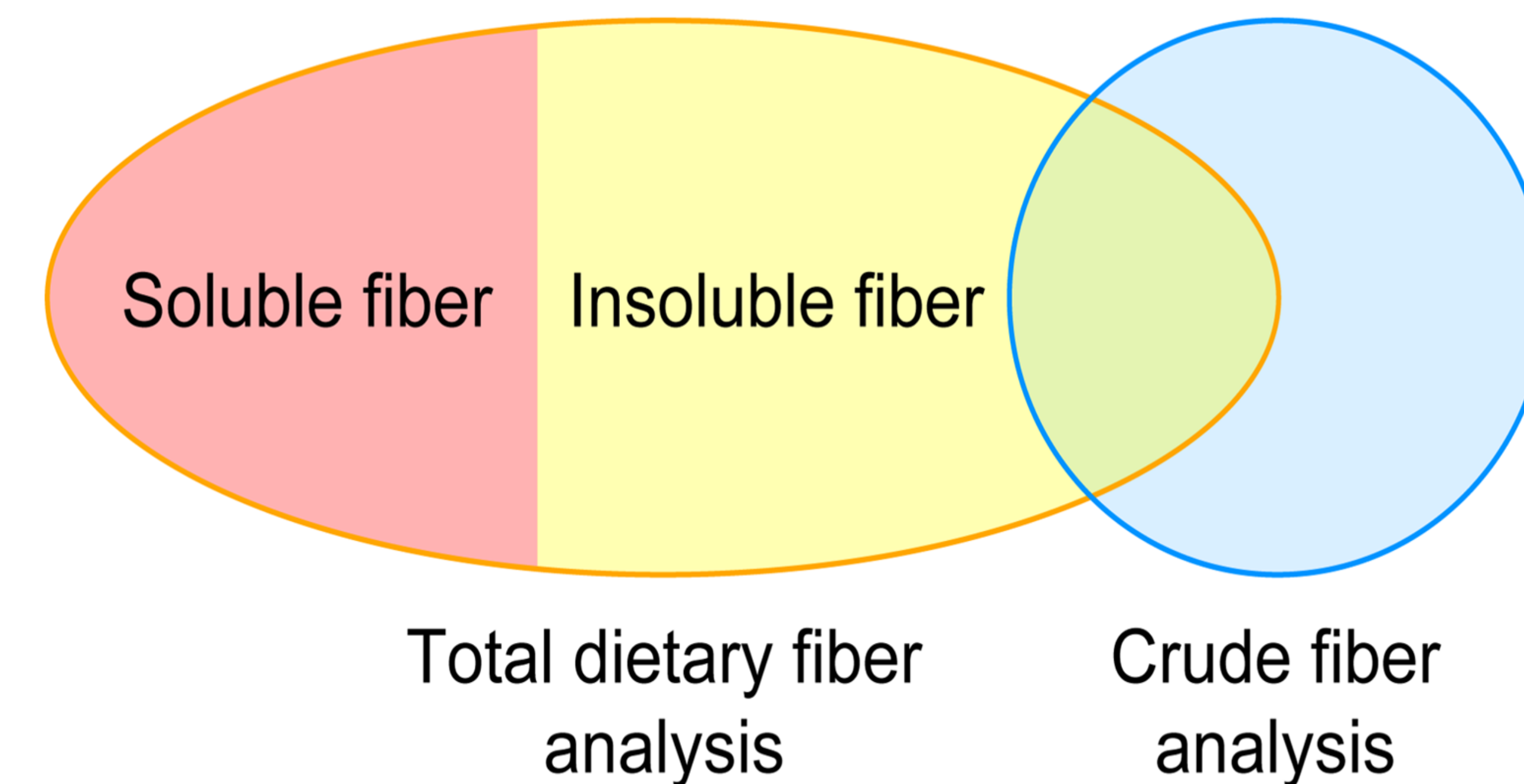
- Dietary fiber is currently not considered an essential nutrient
- AAFCO has no minimum requirement for fiber in either dogs or cats, and carbohydrate-free diets have been fed to dogs short-term without adverse effects
- Some fiber types can reduce mineral absorption, dilute caloric content, and if in excess, cause diarrhea
- There are no published studies documenting this occurring on commercially, balanced diets for dogs and cats



Fiber Reporting



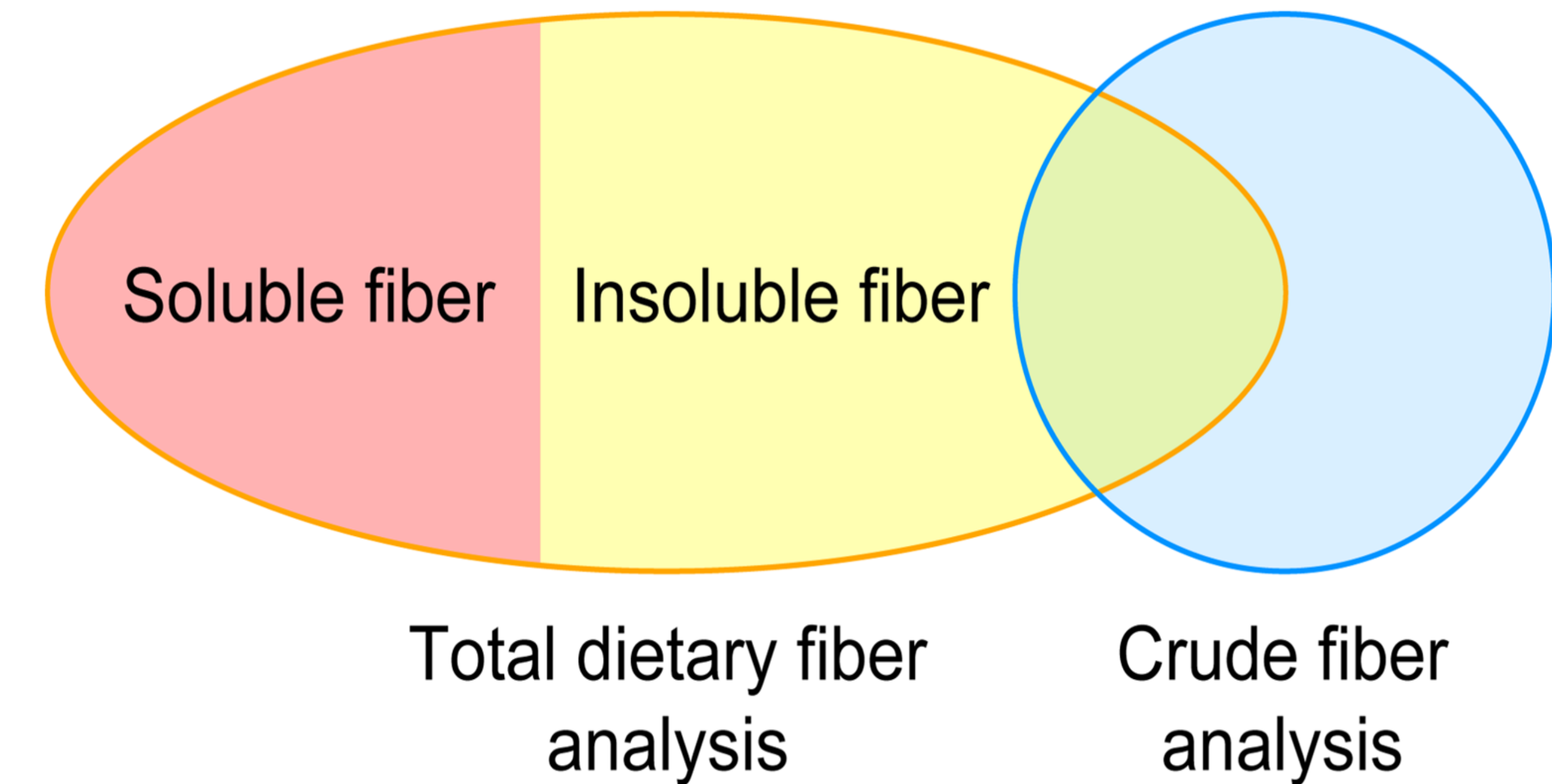
- The most common laboratory method for reporting fiber in pet foods is the crude fiber method
 - AAFCO requires that the guaranteed analysis of pet foods list the maximum crude fiber contents
 - Crude fiber accurately depicts total fiber content in a food only when most of the fiber is slowly fermentable and insoluble
 - Crude fiber does not provide an estimate of highly fermentable and soluble fiber content



Fiber Reporting



- Total dietary fiber is more commonly used in human foods using enzymatic and gravimetric processes
 - This method separates fibers into soluble and non-soluble therefore including the fiber contribution missed with crude analysis



Breaking Up with % as Fed

Purina Beyond Small Dog White Meat
Chicken, Barley and Egg

- 549 kcal/cup
- 4,245 kcal/kg
- Fat:
 - 19.4% as fed
 - 4.6 g/100 kcal

Purina ONE SmartBlend Tender Cutes in
Gravy Chicken and Brown Rice

- 342 kcal/can
- 929 kcal/kg
- Fat Content:
 - 4.4% as fed
 - 4.7 g/100kcal



Step 1: How many
kcal per g of diet?

=1000 g/929 kcal

Step 2: How many g
per 100 kcals of diet?

= 107.64 g /100 kcal

Step 3: What percent
of those g are fat?

= 4.7 g /100 kcal



Too much?

- Tufts Clinical Nutrition Service
 - Nutrient Level Calculator
 - What do you need?
 - How many calories are in your food or treat, in kcal/kg
 - The amount of the nutrient(s) you are interested in (sodium, fat, copper, phosphorus, etc) on an “% as fed” basis



Pet Food Calculator: Comparing nutrient levels between two pet foods



by Clinical Nutrition Team

https://vetnutrition.tufts.edu/2017/08/nutrient_converter/





What is a Fiber Diet?



Dietary Management

Fat Content

Digestibility

Hydrolyzed

Fiber Content

Novel Protein

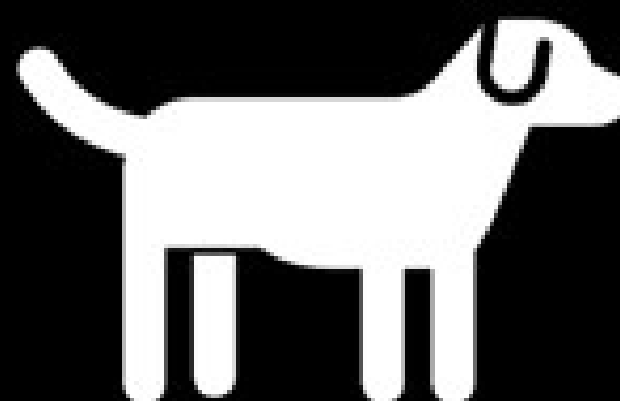


Fiber-Supplemented Diets

- ❑ Fiber is added to diets for multiple reasons and therapeutic indications
 - The fiber type and source will influence the type of effect that is seen in the patient
- ❑ Gastrointestinal diets
- ❑ Diabetic diets
- ❑ Weight management diets



US Commercial Diet Range

Dietary fiber g/100 kcal				
	Crude fiber	TDF	Crude fiber	TDF
Low	< 1.0	< 2.0	< 0.5	< 0.5
Moderate	1.0 - 2.0	2.0 – 4.0	0.5 – 1.0	1.0 – 2.5
High	> 2.0	> 4.0	> 1.0	> 2.5
Range	0.1 – 6.8	0.1 – 11.0	0.1 – 4.9	0.2 – 7.5

Moreno A, et al. 2022.



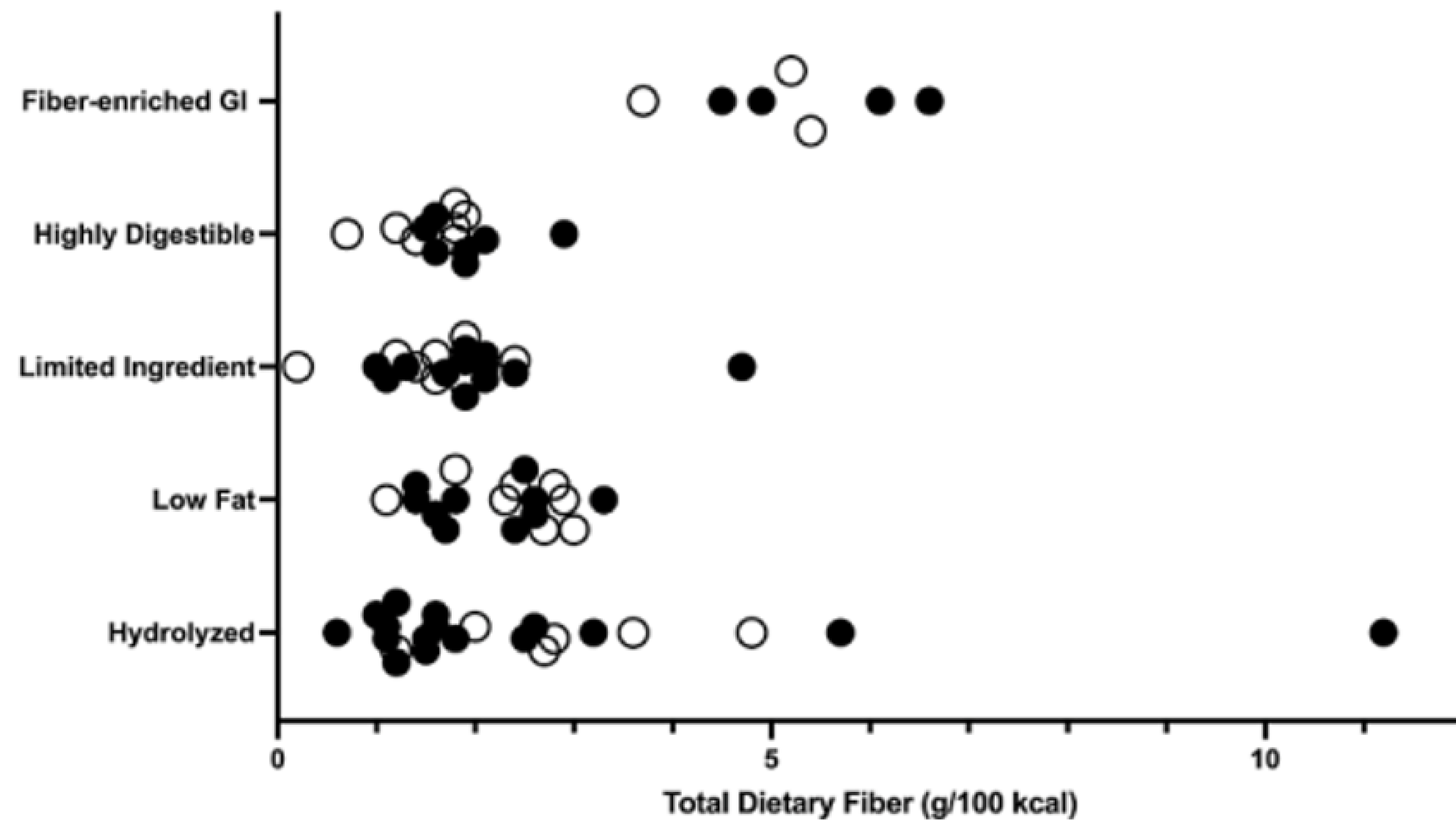
INCREASED FIBER DIETS

Canine (In order of decreasing total dietary fiber concentration; TDF)

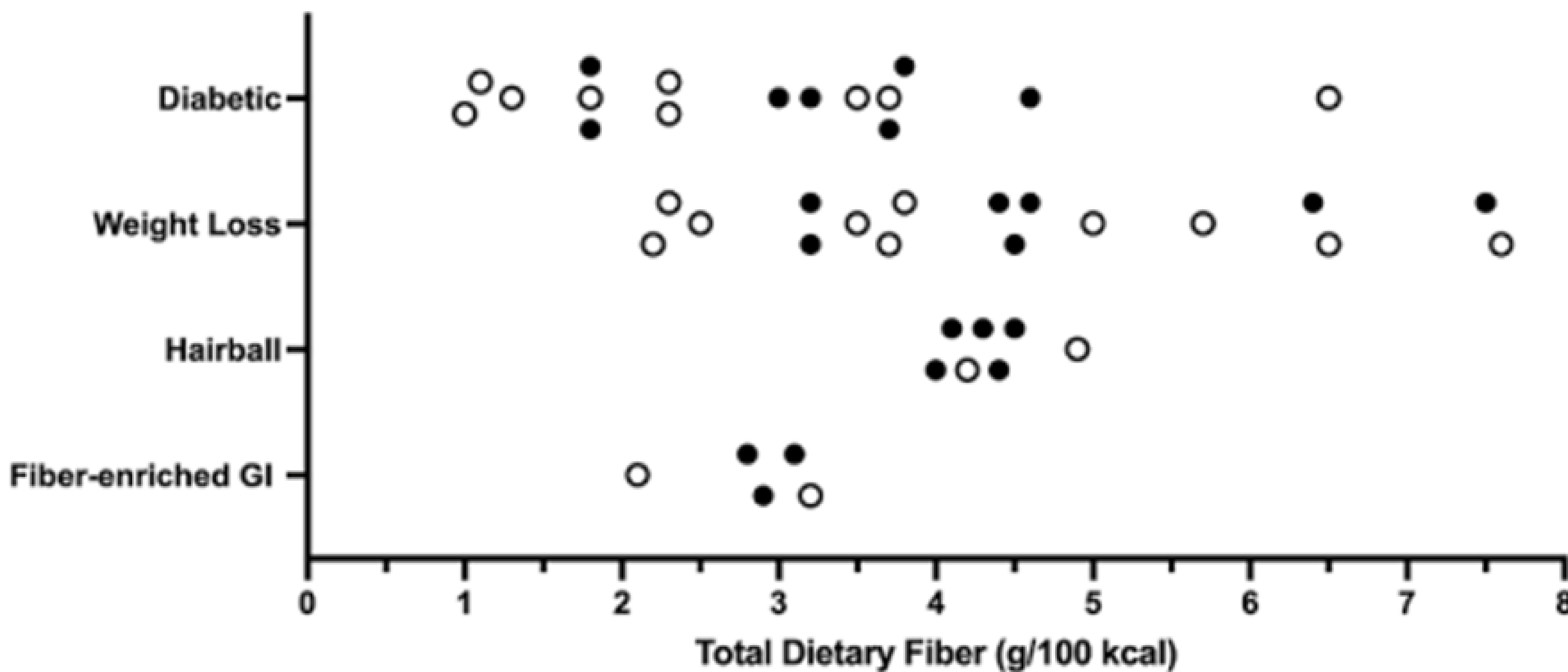
Diet	Kcal per cup/can	Kcal per gram	Nutrients of Concern (/100 kcal)				
			TDF (g)	Crude fiber (g)	Prot (g)	Fat (g)	Phos (mg)
Royal Canin Satiety Support (dry)	214	2.887	10.8	6.4	11.5	3.6	270
Royal Canin Advanced Mobility Support + Satiety (dry)	246	2.997	10.3	5.7	10.2	4.2	150
Royal Canin Satiety Support Small Dog (dry)	228	2.925	10.2	5.3	11.0	3.5	260
Purina Pro Plan Veterinary Diets OM (13.3 oz)	245	0.652	9.6	5.9	13.2	3.6	350
Purina Pro Plan Veterinary Diets OM Select Blend Chicken Flavor (dry)	218	2.929	8.9	3.5	10.0	2.1	280
Hill's Prescription Diet w/d (dry)	255	3.100	8.1	4.7	6.1	3.8	193
Hill's Prescription Diet Metabolic + Mobility (dry)	291	3.210	7.9	4.3	8.0	4.2	201
BLUE Natural Veterinary Diet W+U (12.5 oz)	312	0.881	7.8	4.0	10.2	4.9	350
Hill's Prescription Diet Metabolic (dry)	260	3.157	7.8	3.7	8.2	3.6	215
Hill's Prescription Diet r/d (dry)	245	2.982	7.8	4.1	10.0	2.5	178
Hill's Science Diet Adult Perfect Weight Chicken & Vegetable Entrée (12.8 oz)	310	0.853	7.7	3.4	8.0	3.6	181
Hill's Science Diet Adult Light (regular, small bites or large breed; dry)	271	2.987	7.6	3.8	7.7	2.5	210
Hill's Prescription Diet w/d (13 oz)	307	0.830	7.5	4.0	5.6	3.4	155
Purina Pro Plan Veterinary Diets OM (dry)	256	2.948	7.5	3.6	9.3	2.2	280
Hill's Prescription Diet r/d (12.3 oz)	299	0.854	7.4	3.8	10.1	2.6	177
Royal Canin Large Weight Care (dry)	261	3.182	7.4	3.4	8.7	3.5	190
Royal Canin Medium Weight Care (dry)	256	3.200	7.4	3.4	8.7	3.5	190
Hill's Prescription Diet w/d Vegetable & Chicken Stew (5.5 or 12.5 oz)	89; 201	0.569	7.1	4.4	6.6	3.1	183
Hill's Prescription Diet Metabolic (13 oz)	322	0.861	6.9	4.6	8.4	4.0	177
Hill's Prescription Diet Metabolic Lamb Meal & Rice Formula (dry)	256	3.116	6.9	3.9	7.9	3.5	222
Hill's Science Diet Adult Perfect Weight (dry)	299	3.191	6.6	3.0	8.2	3.2	211
Hill's Science Diet Adult Perfect Weight Large Breed (dry)	289	3.191	6.6	3.0	8.3	3.2	215
Hill's Science Diet Adult Perfect Weight Small Bites (dry)	317	3.191	6.6	3.0	8.3	3.2	215
BLUE Natural Veterinary Diet W+U (dry)	309	3.071	6.5	4.4	10.1	3.6	240
Hill's Science Diet Adult Perfect Weight Hearty Vegetable & Chicken Stew (12.5 oz)	247	0.698	6.5	3.4	8.3	3.5	164
Royal Canin Gastrointestinal High Fiber (dry)	290	3.373	6.3	3.2	7.1	4.9	250
Royal Canin Satiety Support Loaf (13.4 oz)	225	0.592	6.2	4.5	12.3	3.7	230
Royal Canin Urinary SO + Satiety Loaf in Sauce (13.5 oz)	219	0.570	6.0	0.5	13.2	4.1	260
Purina Pro Plan Veterinary Diets EN Fiber Balance (dry)	324	3.385	5.8	1.8	6.5	3.5	240
Royal Canin Small Weight Care (dry)	259	3.273	5.8	2.4	9.1	3.3	190
BLUE Natural Veterinary Diet W+M (dry)	325	3.234	5.6	3.7	9.5	4.1	210
Hill's Prescription Diet Gastrointestinal Biome Chicken & Vegetable Stew (12.5 oz)	286	0.808	5.5	2.0	6.8	4.1	154
Royal Canin Glycobalance Loaf in Sauce (13.4 oz)	286	0.753	5.5	4.3	13.0	5.8	220
BLUE Natural Veterinary Diet W+M (12.5 oz)	331	0.933	5.2	3.0	9.4	4.5	190
Royal Canin Glycobalance (dry)	307	3.334	5.1	2.4	11.0	3.6	210
Royal Canin Weight Control Large Dog (dry)	270	3.287	5.1	2.1	8.4	3.3	180
Hill's Prescription Diet Gastrointestinal Biome Small Bites (dry)	330	3.330	4.7	2.0	5.8	3.5	187
Hill's Prescription Diet Gastrointestinal Biome (dry)	343	3.456	4.5	1.9	5.6	3.3	180
AAFCO minimum	--	--	--	--	4.5	1.4	100



Variability!



Rudinsky AJ, et al. 2026.





Do I Have to Use a Fiber Diet?



KEY POINTS

“Diets manufactured by reputable pet food companies are the ideal method of fiber supplementation for dogs and cats because of their guaranteed analysis and assurance they are complete and balanced.”



LIMITED ANTIGEN / HYDROLYZED (& other “Derm”) DIETS
Canine (In order of increasing fat concentration)

Diet	Kcal per cup/can	Kcal per gram	Protein (g)	Nutrients of Concern (/100 kcal)			
				Fat (g)	TDF (g)	Phos (mg)	Copper (mg)
BLUE Natural Veterinary Diet HF (12.5 oz)	367	1.035	10.4	2.5	1.2	280	1.18
Purina Pro Plan Veterinary Diets HA Vegetarian (dry)	314	3.695	5.4	2.7	1.2	230	0.44
Purina Pro Plan Veterinary Diets EL (dry)	412	3.322	6.3	2.8	0.8	290	0.45
Royal Canin Vegetarian (dry)	289	3.395	5.4	2.8	3.2	150	0.42
Royal Canin Selected Protein PW Moderate Calorie (dry)	257	3.139	7.7	3.0	4.7	240	0.47
Purina Pro Plan Veterinary Diets HA Chicken Flavor (dry)	342	3.782	5.3	3.2	1.1	220	0.42
Royal Canin Hydrolyzed Protein Moderate Calorie (dry)	323	3.488	6.2	3.2	1.4	160	0.43
Royal Canin Hydrolyzed Protein PS (dry)	302	3.516	5.6	3.2	2.0	160	0.40
Royal Canin Selected Protein PD (dry)	288	3.516	5.6	3.2	1.9	190	0.40
Royal Canin Selected Protein PR (dry)	289	3.526	5.6	3.2	1.9	160	0.40
Royal Canin Selected Protein PV (dry)	285	3.477	5.7	3.2	2.0	240	0.41
Royal Canin Selected Protein PW Large Dog (dry)	334	3.477	5.7	3.2	1.9	190	0.41
Purina Pro Plan Veterinary Diets HA Salmon Flavor (dry)	316	3.506	6.7	3.3	1.0	290	0.41
Royal Canin Vegetarian Loaf in Sauce (13.6 oz)	341	0.886	7.9	3.4	1.6	170	0.41
Hill's Prescription Diet z/d (13 oz)	352	0.950	5.2	3.6	2.9	140	0.29
Purina Pro Plan Veterinary Diets HA Chicken Flavor (13.3 oz)	341	0.905	7.7	3.6	4.8	380	0.56
Royal Canin Hydrolyzed Protein Loaf (13.8 oz)	390	0.991	5.9	3.6	2.0	170	0.38
BLUE Natural Veterinary Diet HF (dry)	368	3.529	7.2	3.7	2.5	190	0.38
Hill's Prescription Diet z/d regular or small bites (dry)	354; 374	3.569	4.9	3.7	1.6	150	0.38
Royal Canin Satiety + Hydrolyzed Protein (dry)	208	2.852	11.7	3.7	11.0	230	0.59
Royal Canin Selected Protein PW (dry)	313	3.518	6.5	3.8	2.4	220	0.41
Hill's Prescription Diet d/d Potato & Duck Formula (dry)	373	3.651	4.7	4.0	2.1	145	0.45
Hill's Prescription Diet d/d Potato & Salmon Formula (dry)	372	3.643	4.7	4.0	2.1	164	0.49
Hill's Prescription Diet d/d Potato & Venison Formula (dry)	371	3.636	4.7	4.0	2.1	154	0.44
Hill's Prescription Diet d/d Salmon Formula (13 oz)	405	1.094	4.7	4.0	1.6	170	0.44
Royal Canin Urinary SO + Hydrolyzed Protein (dry)	327	3.592	5.3	4.0	1.8	130	0.40
Hill's Prescription Diet d/d Duck Formula (13 oz)	387	1.046	4.3	4.1	1.6	154	0.52
Royal Canin Hydrolyzed Protein Small Dog (dry)	376	3.653	6.2	4.1	1.1	170	0.41
Royal Canin Skin Support (dry)	322	3.621	6.5	4.1	1.9	260	0.40
Hill's Prescription Diet Derm Complete (13 oz)	370	1.211	4.9	4.1	1.2	134	0.56
Hill's Prescription Diet Derm Complete (dry)	388	3.799	4.2	4.2	1.0	132	0.28
Royal Canin Ultamino (dry)	323	3.755	4.6	4.2	1.5	120	0.38
Royal Canin Selected Protein PW Loaf (13.6 oz)	414	1.076	6.4	4.3	2.1	250	0.33
Purina Pro Plan Adult Sensitive Skin & Stomach Salmon & Rice (dry)	467	4.049	6.7	4.3	--	279	0.37
Purina Pro Plan Veterinary Diets DRM Naturals (dry)	418	3.871	6.5	4.4	2.4	250	0.41
Royal Canin Selected Protein PV Loaf (13.6 oz)	398	1.034	6.3	4.5	2.0	180	0.81
Royal Canin Selected Protein PR Loaf (13.6 oz)	416	1.080	6.2	4.7	1.8	170	0.37
Royal Canin Hydrolyzed Protein HP (dry)	332	3.856	5.3	4.8	1.5	210	0.25
Royal Canin Selected Protein PD Loaf (13.6 oz)	429	1.113	5.8	5.2	1.7	180	0.39
AAFCO minimum		---	4.5	1.4	--	100	0.18



Fibre Supplementation

- Proven benefit in chronic colitis cases
- SCFA production
- Enhance structure and function of the epithelium
- Colonic motility
- Dosing per meal
 - 0.5-1 T for small dogs
 - 2 T for medium dogs
 - 3T for large dogs



KEY POINTS

“In the one study reporting fiber dosing, dogs with fiber responsive large bowel disease reportedly responded to a median dosage of 2 tablespoons per day psyllium fiber.”

“The full range of fiber supplementation used in that study was between 0.25 to 6 tablespoons per day.”

Leib MS, et al. 2000.



Fibre Supplementation

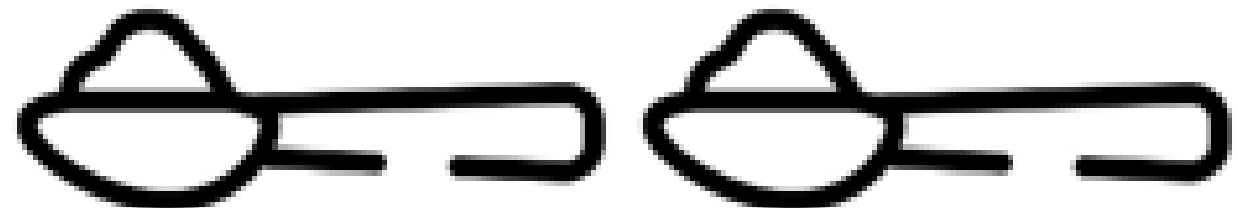
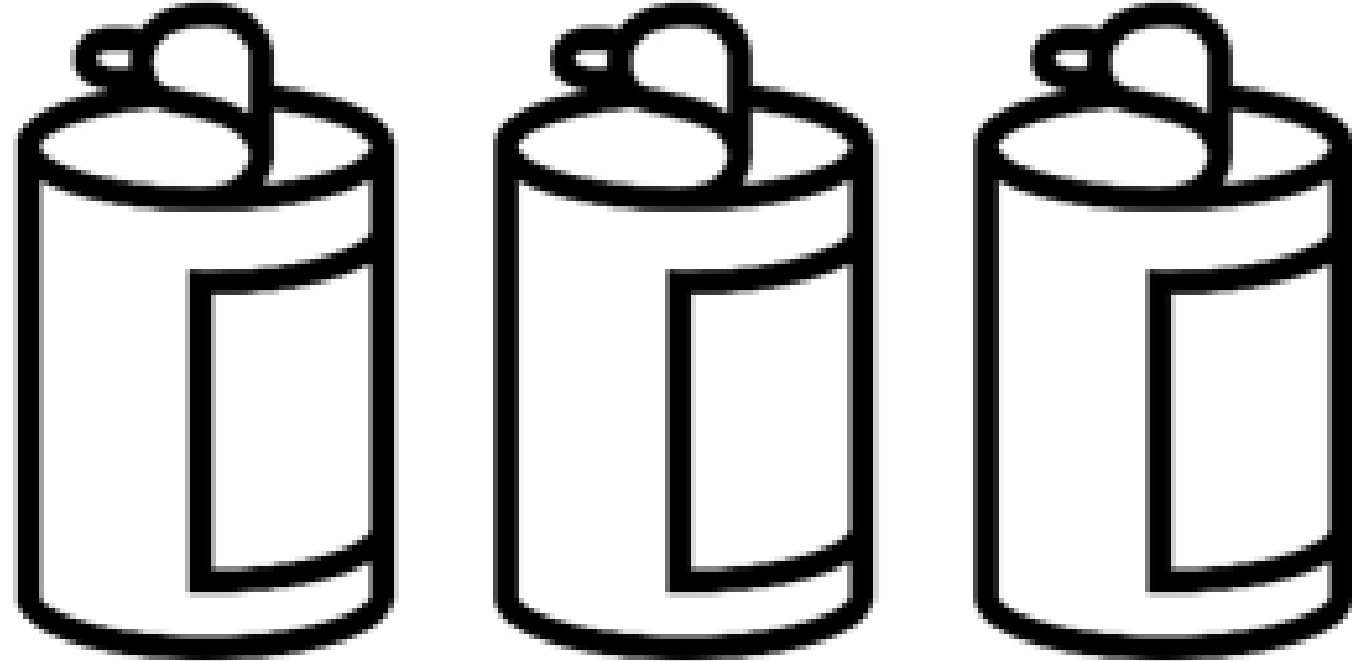
- Proven benefit in chronic colitis cases
- Fermentable CHO's -> SCFA production -> Energy for colonocytes
 - Psyllium
- Enhance structure and function of the epithelium
- Colonic motility

Dosing per meal

- 0.5-1 T for small dogs
- 2 T for medium dogs
- 3T for large dogs



Moreno A, et al. 2022.

Fiber Source	Amount	Total Fiber ~20 grams
Psyllium husk	2 tbsp	
Canned pumpkin	3 cups	
Green beans	8 cups	



Fiber Fail?



SUMMARY

- Fiber is a complex component of canine and feline nutrition.
- Understanding the basics of fiber is useful to practicing clinicians.
- Specific types or amounts of fiber are needed for specific clinical effects.
- Canine and feline gastrointestinal disease can often be effectively managed with fiber-enhanced diets and in many cases offers many advantages over alternative treatment options.





THANK YOU

